

## VYBORG, Russian Federation

WMO: 228920

Lat: 60.717N

Lon: 28.733E

Elev: 43

StdP: 14.67

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -10.7      | -5.4 | -15.5                         | 2.4 | -10.7 | -9.9 | 3.2 | -5.2 | 27.6                  | 31.7 | 25.7 | 31.6 | 2.4                   | 0    | 0.618 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.6                   | 81.0            | 67.0 | 77.5 | 65.3 | 74.5 | 63.7 | 69.2                | 77.5 | 67.2 | 75.1 | 65.3 | 72.5 | 6.8                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.1                            | 96.4 | 74.2 | 64.0 | 89.4 | 71.8 | 62.2 | 83.8 | 69.6 | 33.3          | 77.5 | 31.7 | 75.3 | 30.2 | 72.7 | 76.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 24.3          | 21.0 | 18.3 | DB                         | -14.0 | 84.8               | 8.0 | 4.0                                                | -19.8 | 87.6       | -24.4 | 90.0       | -28.9 | 92.2       | -34.7 | 95.1 | (4) |
| (5)               |      |      | WB                         | -14.1 | 71.3               | 7.8 | 2.5                                                | -19.7 | 73.1       | -24.3 | 74.6       | -28.6 | 76.0       | -34.3 | 77.8 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 41.5  | 20.4  | 19.8  | 26.7  | 38.0  | 50.5  | 59.4  | 65.4  | 62.2  | 53.0  | 42.1  | 33.0  | 25.8  | (6)  |
| (7)  |                                                                   | DBStd       | 17.78 | 12.15 | 11.99 | 8.19  | 6.53  | 7.45  | 6.30  | 5.72  | 5.34  | 5.89  | 6.74  | 7.55  | 10.55 | (7)  |
| (8)  |                                                                   | HDD50       | 4487  | 916   | 844   | 722   | 365   | 85    | 4     | 0     | 0     | 37    | 254   | 511   | 749   | (8)  |
| (9)  |                                                                   | HDD65       | 8714  | 1381  | 1264  | 1187  | 811   | 453   | 191   | 65    | 118   | 359   | 710   | 961   | 1214  | (9)  |
| (10) |                                                                   | CDD50       | 1386  | 0     | 0     | 0     | 4     | 100   | 286   | 478   | 380   | 129   | 9     | 0     | 0     | (10) |
| (11) |                                                                   | CDD65       | 134   | 0     | 0     | 0     | 0     | 3     | 22    | 77    | 32    | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                   | CDH74       | 772   | 0     | 0     | 0     | 0     | 21    | 137   | 456   | 158   | 0     | 0     | 0     | 0     | (12) |
| (13) |                                                                   | CDH80       | 132   | 0     | 0     | 0     | 0     | 1     | 21    | 89    | 21    | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                              | WSAvg       | 7.5   | 7.7   | 7.5   | 7.3   | 6.5   | 7.4   | 7.7   | 6.8   | 6.7   | 7.2   | 7.8   | 8.4   | 8.7   | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 27.3  | 1.8   | 1.4   | 1.7   | 1.5   | 1.5   | 2.0   | 2.8   | 3.2   | 3.1   | 2.9   | 3.0   | 2.6   | (15) |
| (16) |                                                                   | PrecMax     | 39.9  | 3.0   | 3.5   | 3.9   | 3.8   | 3.5   | 4.7   | 5.7   | 7.4   | 6.3   | 7.2   | 6.9   | 5.0   | (16) |
| (17) |                                                                   | PrecMin     | 19.1  | 0.3   | 0.0   | 0.4   | 0.2   | 0.2   | 0.1   | 0.7   | 0.7   | 1.0   | 0.8   | 0.2   | 0.6   | (17) |
| (18) |                                                                   | PrecStd     | 5.4   | 0.7   | 0.7   | 0.9   | 0.8   | 0.8   | 1.3   | 1.4   | 1.6   | 1.5   | 1.7   | 1.2   | 1.2   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 38.1  | 38.7  | 46.2  | 63.7  | 76.7  | 82.8  | 86.6  | 82.7  | 70.6  | 57.4  | 48.3  | 44.9  | (19) |
| (20) |                                                                   | MCWB        | 36.8  | 35.5  | 37.7  | 51.2  | 61.9  | 66.8  | 69.8  | 67.7  | 61.2  | 53.4  | 46.3  | 43.2  | (20)  |      |
| (21) |                                                                   | 2%          | DB    | 36.5  | 36.2  | 41.1  | 57.2  | 71.4  | 77.9  | 82.3  | 78.2  | 66.7  | 55.1  | 46.3  | 41.4  | (21) |
| (22) |                                                                   | MCWB        | 35.4  | 34.3  | 35.4  | 47.6  | 58.2  | 64.5  | 68.2  | 65.9  | 59.2  | 52.5  | 44.6  | 40.1  | (22)  |      |
| (23) |                                                                   | 5%          | DB    | 35.1  | 34.9  | 37.8  | 52.2  | 66.9  | 73.8  | 79.0  | 74.9  | 64.2  | 53.2  | 44.7  | 39.5  | (23) |
| (24) |                                                                   | MCWB        | 34.2  | 33.5  | 34.0  | 44.6  | 55.5  | 62.3  | 66.9  | 64.3  | 58.2  | 51.0  | 43.3  | 38.2  | (24)  |      |
| (25) |                                                                   | 10%         | DB    | 33.8  | 33.8  | 35.8  | 48.3  | 62.9  | 70.1  | 75.8  | 71.8  | 61.9  | 51.4  | 42.6  | 37.4  | (25) |
| (26) |                                                                   | MCWB        | 32.9  | 32.8  | 33.2  | 41.5  | 53.2  | 60.1  | 65.0  | 63.0  | 56.8  | 49.5  | 41.3  | 36.3  | (26)  |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 37.4  | 36.5  | 39.3  | 53.0  | 64.0  | 69.8  | 72.5  | 70.7  | 63.1  | 54.8  | 46.8  | 43.2  | (27) |
| (28) |                                                                   | MDCB        | 37.7  | 38.0  | 43.6  | 60.9  | 72.3  | 78.9  | 82.8  | 77.6  | 67.8  | 56.3  | 47.7  | 44.5  | (28)  |      |
| (29) |                                                                   | 2%          | WB    | 35.6  | 34.9  | 36.4  | 48.8  | 60.0  | 67.0  | 70.1  | 67.5  | 61.0  | 53.0  | 44.9  | 40.4  | (29) |
| (30) |                                                                   | MDCB        | 36.2  | 35.8  | 39.3  | 56.2  | 68.6  | 75.1  | 79.0  | 75.4  | 64.6  | 54.5  | 45.9  | 41.2  | (30)  |      |
| (31) |                                                                   | 5%          | WB    | 34.3  | 33.9  | 34.8  | 45.0  | 57.1  | 64.1  | 68.3  | 65.8  | 59.3  | 51.3  | 43.3  | 38.3  | (31) |
| (32) |                                                                   | MDCB        | 34.9  | 34.6  | 36.8  | 50.7  | 65.4  | 70.9  | 76.3  | 73.1  | 63.0  | 52.8  | 44.4  | 39.3  | (32)  |      |
| (33) |                                                                   | 10%         | WB    | 33.1  | 32.8  | 33.8  | 42.4  | 54.1  | 61.5  | 66.4  | 64.0  | 57.7  | 49.8  | 41.5  | 36.4  | (33) |
| (34) |                                                                   | MDCB        | 33.7  | 33.7  | 35.3  | 47.2  | 61.3  | 68.4  | 74.0  | 70.0  | 61.1  | 51.2  | 42.5  | 37.2  | (34)  |      |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 8.8   | 10.3  | 11.8  | 12.7  | 13.8  | 12.5  | 12.6  | 10.8  | 7.6   | 6.1   | 7.3   | (35)  |      |
| (36) |                                                                   | 5% DB       | MCDBR | 5.7   | 7.2   | 12.6  | 20.1  | 20.2  | 18.3  | 17.6  | 17.5  | 13.8  | 8.3   | 6.5   | 5.7   | (36) |
| (37) |                                                                   |             | MCWBR | 5.8   | 6.4   | 9.0   | 12.1  | 10.4  | 8.5   | 7.3   | 7.9   | 8.0   | 6.4   | 6.5   | 5.6   | (37) |
| (38) |                                                                   |             | 5% WB | MCDBR | 5.8   | 6.9   | 9.6   | 18.1  | 17.6  | 15.3  | 15.2  | 14.7  | 11.5  | 7.3   | 6.0   | 5.4  |
| (39) |                                                                   | MCWBR       |       | 6.0   | 6.4   | 7.4   | 11.4  | 10.3  | 8.4   | 7.3   | 7.9   | 7.6   | 6.3   | 6.2   | 5.5   | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.267 | 0.306 | 0.329 | 0.368 | 0.364 | 0.362 | 0.387 | 0.376 | 0.350 | 0.332 | 0.294 | 0.247 | (40)  |      |
| (41) |                                                                   | taud        | 2.089 | 2.122 | 2.181 | 2.296 | 2.414 | 2.457 | 2.406 | 2.429 | 2.490 | 2.385 | 2.180 | 2.042 | (41)  |      |
| (42) |                                                                   | Ebn at Noon | 145   | 209   | 248   | 261   | 273   | 275   | 264   | 258   | 245   | 205   | 134   | 100   | (42)  |      |
| (43) |                                                                   | Edh at Noon | 13    | 24    | 31    | 34    | 33    | 32    | 33    | 30    | 23    | 19    | 12    | 9     | (43)  |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 79    | 261   | 725   | 1183  | 1626  | 1702  | 1670  | 1287  | 761   | 306   | 82    | 41    | (44)  |      |
| (45) |                                                                   | RadStd      | 16    | 58    | 93    | 128   | 187   | 115   | 154   | 143   | 80    | 42    | 13    | 5     | (45)  |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |      |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |      |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |      |
| (46) | Station Only            | +0.92   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | N/A | (46) |
| (47) | Regional (31 neighbors) | +0.97   | N/A    | N/A     | N/A   | N/A   | N/A         | -329  | -340  | N/A   | N/A | (47) |

Nomenclature: See separate page